

Balancing Risk and Costs to Optimize the Clinical Supply Chain—A Step Beyond Simulation

Chedia Abdelkafi · Benoît H. L. Beck · Benoit David ·
Cédric Druck · Mitchell Horoho

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Abstract With increasing pressure to accelerate drug development and minimize associated costs, it has become critical for pharmaceutical companies to optimize the clinical supply chain. Various tools have been developed to improve forecasts of medication requirements, some of them based on Monte Carlo simulation techniques. In this paper, we describe an innovative approach that goes beyond simulating trials with a priori supply strategies. This approach optimizes the supply plan by balancing the various costs against the risk of running out of medication and utilizes the Bayesian principle to reevaluate supply strategies over time. Supporting methodologies and processes, key to a successful implementation, are also emphasized.

Keywords Clinical supply chain optimization · Bayesian · Decision modeling · Monte Carlo · Simulation · Cost · Risk

Introduction

With increasing pressure to accelerate drug development and minimize associated costs, it has become critical for pharmaceutical companies to optimize the clinical trial supply chain. Various tools have, as a consequence, been developed to improve forecasts of medication requirements, some of them based on Monte Carlo simulation techniques.

In this paper, we describe an innovative approach that goes beyond simulating trials with a priori supply strategies. This approach optimizes the supply plan by balancing the various costs (i.e., manufacturing, packaging, distribution, etc.) against the risk of running out of medication and utilizes the Bayesian principle to reevaluate supply strategies over time.

Supporting methodologies and processes, key to a successful implementation, are also emphasized.

Background

Balancing Risk and Cost in the Clinical Supply Chain

The clinical trial supply chain entails much specificity compared to standard supply chains.

Constraints like expiration dating, bulk availability, and specific country labeling must be taken into account. In addition, trial designs (e.g., stratification, randomization, titration), patient enrollment, dropouts, and drug distribution are factors that generate significant uncertainty in the forecast of material needs.

This overall uncertainty in the demand forecast leads to some risk of being unable to supply the right drug to the

C. Abdelkafi (✉)
Clinical Supplies Solutions, N-SIDE LLC,
3711 Market Street,
Philadelphia, PA 19104, USA
e-mail: chedia.abdelkafi@n-side.com

B. H. L. Beck
Quantitative Modeling and Decision Sciences,
AXIOSIS, sprl, Drève Emmanuelle 28A,
1470 Bousval, Belgium

B. David
Operations Research Consulting, N-SIDE SA,
Chemin du Cyclotron, 6,
1348 Louvain-La-Neuve, Belgium

M. Horoho
Clinical Trial Materials Services, Eli Lilly and Company,
Lilly Corporate Center, DC 3911,
Indianapolis, IN 46285, USA

C. Druck
Clinical Supplies Solutions, N-SIDE SA,
Chemin du Cyclotron, 6,
1348 Louvain-La-Neuve, Belgium

right patient at the right time. In order to reduce this risk, various techniques can be used, like increased overage, increased shipment frequency, dynamic supply rules (e.g., shipping to site after patient randomization), and frequent real-time inventory tracking.

As each of these techniques implies a certain cost, the goal of a clinical supply organization is to find the right balance between these costs and risk for each trial.

Because of the economical context, it is no longer sustainable to allocate huge amounts of overage to manufactured batches, and this is even less sustainable with the expensive and complex-to-produce biologics that represent an increasing portion of new drugs being developed. However, focusing solely on “waste minimization” is not a good practice either, as it prevents a comprehensive analysis of the different costs relative to clinical supplies. A full cost optimization is yet difficult to implement in practice, as this would require the translation of risk into cost despite its intangible aspect.

Therefore, the actual objective function that must be solved can be described as:

Minimize Cost (product, operations, distribution, human resources, opportunity) by Risk Level

Where:

- Product cost = material costs
- Operations cost = manufacturing and packaging fixed and variable costs
- Distribution cost = shipment and storage costs
- Human resources cost = cost of time spent by employees on supplies management
- Opportunity cost = opportunity cost of capital immobilized in resources and inventory
- Risk level = probability of being unable to supply the appropriate drug for a patient visit

The result is represented in Fig. 1, where the blue curve corresponds to the set of minimum costs per risk level, and the black dots are all the possible supply strategies that do not correspond to an optimal solution.

The selection of the optimal strategy must be made based on standard decision criteria, taking into account the type of trial (e.g., ABC classification [1]) and therefore, the type of tradeoffs that must be made with regard to risk and cost.

Indeed, in the case, for instance, of a clinical trial with expensive and scarce medication, more emphasis will be placed on reducing the overage, which means a higher risk will have to be accepted (we will see later in this paper that some of the risk that is measured before trial start can be addressed through reevaluation using actual data).

Conversely, if trial medication can be produced in large quantities at a relatively low cost, it can be appropriate to

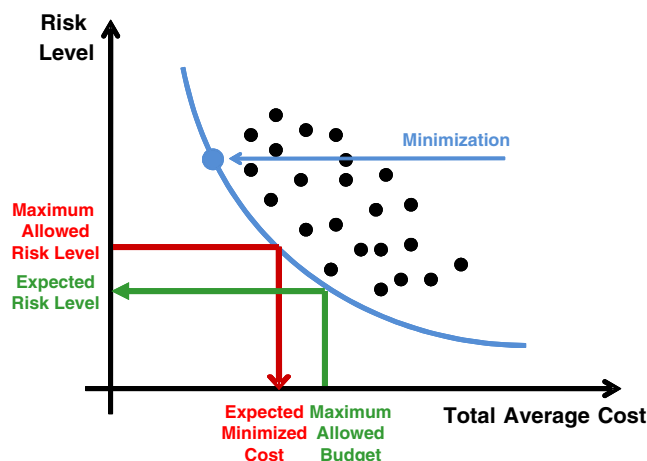


Fig. 1 Graphical representation of the conditional objective function in clinical supplies optimization

increase the overage in order to reduce risk, as well as the overall cost which also includes distribution and inventory tracking resources.

Managing the Clinical Supply Chain

Clinical supplies management is based partially on standard supply chain techniques like economic batch quantity (EBQ), economic order quantity (EOQ), and reorder levels/points (ROL/ROP) [1]. Because of the specific constraints and the inherent variability characterizing clinical supplies, these techniques cannot be applied without more sophisticated approaches and the use of supporting technology.

Interactive Voice Response (IVR) Systems [2] have been used for many years to help manage the clinical supply chain in an automated fashion. Two types of automated inventory management methods are typically used in IVR systems, with some slight differences between the various systems. In practice, project managers—either directly at pharmaceutical companies or through Contract Research Organizations—must determine values for the following parameters in the IVR system:

- Trigger-based method: *Minimum and Maximum stock levels* per package type and location
- Predictive method: *Prediction window* and in some cases *Safety stock level* per package type and location.

The values allocated to these parameters will have a direct impact on shipment frequency and volume, on overage, as well as on the risk to run out of drug. More specifically:

- The minimum/safety stock level will influence the number of stockouts: this quantity must cover the maximum potential demand over the shipping lead time.

- The maximum stock level (or the prediction window) will influence shipment frequency.
- Both will have an impact on overage for the trial.

For instance, when the maximum stock level at a site is increased (with a constant minimum level), a resupply will be required less frequently, and this will, therefore, generate less frequent shipments. Another instance is when increasing the prediction window with a predictive method: the forecasted patient needs will increase, as well as the quantity shipped; therefore, the shipment frequency will decrease. In that case, because of the increased associated uncertainty, the required overage is likely to increase significantly.

These parameters should, therefore, be selected carefully in conjunction with the determination of an optimal production plan to achieve the desired balance between cost and risk.

Forecasting Technology

Various tools have been developed over the past few years to improve forecasts of medication requirements. Many of these tools rely on deterministic calculation. The added value of deterministic tools is significantly limited. These do not permit any method to evaluate the risk associated to a supply strategy.

More advanced tools are typically based on Monte Carlo simulation techniques [3]. This approach was a significant step towards more sophisticated clinical supply chain management, as it enables to consider risks as they relate to variation in inputs such as enrollment rate, dropout, titration, etc. This allows clinical supply managers to have an idea of the variability inherent to each clinical trial and, therefore, determine what the demand could be as well as the risk associated with different supply strategies.

However, as actual clinical supply management is considered in practice, entering an expected output (i.e., supply strategy) as a simulation input still presents some limitations. The simulation methodology is represented on Fig. 2a. The clinical supply manager defines a priori, from various sources, the trial design, distribution network, production quantities with dates, and IVRS inventory management rules and then uses trial simulation to evaluate the outcome of the strategy. A good analysis tool based on trial simulation may identify gaps in the strategy, allowing the clinical supply manager to alter inputs such as minimum stocking levels for a country and then reevaluate through simulation. This creates a very iterative cycle to determine a preferred supply strategy. Furthermore, this type of approach still lacks an actual optimization, since it is not practically

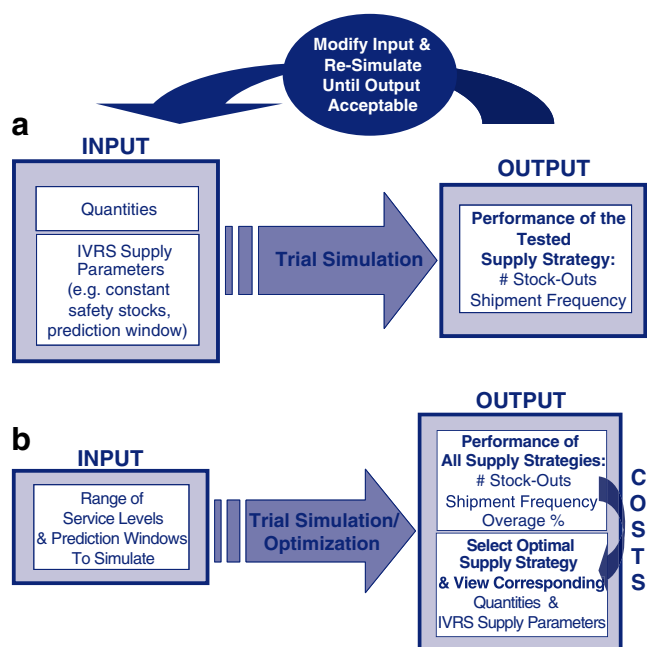


Fig. 2 **a** Typical simulation methodology in forecasting tools. **b** Combined simulation and optimization methodology

possible to manually reach an optimum for all parameters combined.

What clinical supply managers need is a way to determine probable quantities and dates for productions as well as inventory management parameters to enter in their IVRS. For that purpose, an automated optimization combined with simulation can add significant value. The combined methodology, described in detail in the next sections, is represented in Fig. 2b. Multiple supply strategies are simulated - and optimal overages and e.g., safety stocks are calculated - based on the range of service levels and prediction windows entered by the clinical supply manager. The output is a list of performance indicators for each of the simulated strategies. The preferred supply strategy can then be selected based on risk and cost, and the corresponding details (i.e., quantities, safety stocks over time, etc.) can be viewed.

Finally, this technique can also be applied after the clinical trial has started. This allows the reevaluation of the supply strategy as the uncertainty decreases by leveraging actual data collected through the IVR system.

Mid-study simulations using actual data have been mentioned in other publications [3], but to our knowledge, this is the first time a Bayesian approach is taken to reevaluate clinical supply strategies. This approach goes a step further, as it does not only use a snapshot of the situation at a certain time but fully leverage all the data accumulated since the beginning of the trial in order to increase the accuracy of key assumptions like enrollment rates, dropout rates, dose titration probabilities, etc.

Methods

The following subsections, respectively, introduce the design of the proposed solution and the simulation, optimization, and reevaluation aspects that are vital to its implementation. The combination of these three techniques in an efficient and practical way constitutes, from a technical point of view, a novel approach to clinical supplies management.

Model Design

The design of the model is divided into two major parts, i.e., the treatment process and the supply process. The treatment process essentially investigates the possible package demand by site whereas the supply process models the distribution strategy based on levels for the supply management rules automatically selected by the tool.

More specifically, the treatment process models patient enrollment at the investigation sites and the assigned treatment sequences in order to predict package usage quantities for each site. In order to achieve that goal, the treatment process part takes as input—if meaningful for the investigated study—the site opening rate(s), patient enrollment rate(s), stratification ratio(s), weight ranges, visit windows, dropout rate(s), dose titration probabilities, and the randomization scheme. The parameters associated with these characteristics are obtained directly from the study protocol or deduced/estimated from practical characteristics of the trial implementation (countries, populations, etc.)—often known from previous similar clinical investigations.

The supply process model mimics the progress, timeslot by timeslot, of packages through the distribution network. A specific inventory management rule is attached to each node of the network selected from the two types of (re) supply rules introduced in the previous section, i.e., the trigger- and the prediction-based rules.

Simulation Technique

The entire solution relies on a discrete event simulation engine [4]. The implementation is made up of entities (such as packages, patients, countries, sites, etc.) and relationships between these entities (such as shipment of packages, recruitment of patients, package consumption, etc.) that mimic real-life processes occurring during the clinical trial, including the distribution of medication as well as patients' progress through the tested treatments. Such a model is well suited to realistically represent complex dynamic systems from the real world and can easily incorporate stochastic (i.e., random) aspects required for appropriately forecasting trials under uncertainty. Typical trial simulation processes have been described in previous publications [3].

Most of the stochastic aspects are involved in the treatment model part of the solution. They are mathematically represented by random variables with appropriate statistical distributions. Poisson processes are used to model enrollment and site initiation, and discrete probability distributions are used to model stratification, randomization, discontinuation, dose titrations, and weight ranges. Finite range bell-shaped distributions over the visit window (modeled with a Beta-2-2 distribution) are used for obtaining the visit intervals. Multiple Monte Carlo simulations are performed in order to get a large number of realizations for all stochastic processes and robustly investigate all possible outcomes. The goal is to characterize the drug usage forecast on a timeslot basis, as well as estimate the associated uncertainty. These simulations allow for the characterization of the variability levels on the package demands used for selecting inventory management rules (e.g., safety stock levels) satisfying a predefined targeted service level (e.g., 99%). Thousands of independent simulations are typically required to sufficiently cover the overall variability of large studies and to produce robust estimates of the management rule quantities.

Optimization Technique

The selection of the supply plan and inventory management rules is made by automatically investigating several strategies based on different overages and different inventory management values (i.e., trigger levels or prediction windows/safety stocks). In practice, the clinical supply manager is asked to provide some ranges of investigation for key inventory management parameters, like the prediction window and the level of security used to compute safety stocks. These ranges generate a number of scenarios. For each scenario, the supply process model evaluates the ability of the supply strategy to meet the demand and measures some global logistics indicators, like the risk of stockout, the number and volume of shipments, and the overage level.

These indicators are used to find the best balance between the overage (i.e., cost of product/ manufacturing/ packaging), shipments (i.e., cost of distribution), and the risk of shortage. The selection of the best supply scenario is performed by the clinical supply manager according to predefined criteria and any external constraints.

Reevaluation Technique

As stated before, the reevaluation of some of the parameters is critical for improving the precision of the forecast computed by the tool and characterizing the required supply strategy update. In the case of trials with multiple supplies, reevaluations can allow for reductions in waste in later replenishment supply orders. The methodology select-

ed for updating parameter values follows the Bayesian principle [5], i.e., balancing a priori assumptions with actual observations obtained in real-time from the IVRS.

This allows the clinical supply manager to update and correct assumptions (e.g., weight ranges, dose titration probabilities, etc.) throughout the trial in order to reforecast material needs and adjust the inventory management parameters in the IVRS.

In the case of probabilities associated to a discrete stochastic variable, the derivation of the formula for the update of the parameters is based on the Dirichlet or multinomial Bayesian model [5]. The deduced updating formula is essentially a linear combination of the probability levels estimated by the clinical supply manager before study start and of the empirical probability computed from the actual observations in the trial. This combination is weighted by a factor that depends on (1) the number of observations available and (2) the confidence level in the clinical supply manager's assumptions. It is noted that the same update formula can similarly be used for updating or reevaluating stratum probabilities, weight range probabilities, dose titration rates, dropout rates, etc. In practice, clinical supply managers will have to provide, in addition to predicted probability levels, a qualitative measure of confidence in their prediction.

Similarly, a formula for the update of the enrollment rates can also be obtained by following the Bayesian principle. When assuming the homogeneity of the studied Poisson process, the formula simply consists in updating the predicted rate by using the number of occurrences weighted by the observation horizon. In all cases, the clinical supply manager can incorporate any additional information obtained from clinical sites into the prediction.

Case Study—Determining an Optimal Supply Strategy

Clinical Trial Description

Patients are enrolled at 30 investigational sites spread over six countries with a three-layer distribution network as shown in Fig. 3a.

The study plans to randomize a total of 650 subjects over a period of 6 months with different enrollment schemes for each country. The treatment involves seven visits spanning a period of 10 weeks, with an optional follow-up phase (V100 and V101) for discontinued patients. Subjects are stratified based on gender at the study level and randomly assigned to two treatment groups, a placebo and an active one, with a balanced block size of 4. Three dose levels are investigated, and titration between dose levels may occur for the active treatment arm as shown in Fig. 3b.

Two different weight ranges are considered, with country-specific probability distributions. The type and

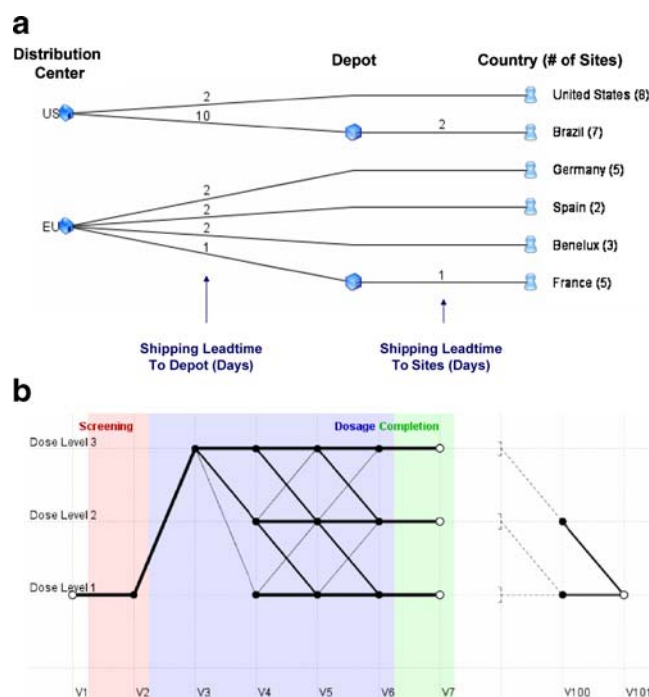


Fig. 3 **a** Distribution network; **b** Visit schedule and dose titration mapping

number of packages dispensed depends on treatment assignment, patient's weight range and target dose level, as well as the visit interval.

Patients receive a total of 16 packages during the main study phase, which spreads from the second visit (i.e., randomization) to the seventh visit, and potentially four additional packages if they are discontinued and choose to enter the taper phase.

Screening failure is expected to be 15% and overall patient dropout rate 10%.

Seven different package types are required to dispense the multiple doses and placebo, and there are two label groups for the study: one single label for the USA and one booklet label for all other countries. This translates into 14 separate stocks that must be managed throughout the supply chain. Product shelf-life is initially 12 months.

In this example, we assume an IVRS predictive inventory management strategy: material needs are forecasted over a certain time horizon, with a frequency equal to the horizon itself. In order to account for the uncertainty, a safety stock is also used and will trigger a reforecast as needed.

Objectives

Ideally, the supply strategy should be initiated at an early stage, when all aspects (i.e., packaging and labeling design, expiration dating, protocol, etc.) can still be influenced.

The first step in creating a supply plan for the trial is to determine what category of trial is addressed in terms of

cost, risk, and other factors that should tell us how much time and resources we want to invest in supply planning and, therefore, in what level of detail we need to go.

We will first consider a prestudy forecast, with the objective of defining optimal supply plan and IVRS inventory management parameters before study start based on the information available at that time.

In the second stage, we will utilize actual data extracted from the IVR system approximately 7 months after the first patient visit to reevaluate assumptions and adjust the supply strategy.

Prestudy Forecast

Simulation/Optimization Design

For the first part of the simulation (i.e., treatment), 5,000 independent realizations of the trial are generated in order to have a fair estimate of the full variability of product needs at each investigational site and for each package type and label group. Based upon this estimate, the inventory management rules are computed, and the supply through the distribution network is, in turn, simulated 1,000 times.

A range of prediction windows (7–112 days) and of confidence levels on the computation of safety stocks (70–99.98%) is investigated. Each combination of prediction window and confidence level represents a supply scenario that is tested through simulation.

We limit the changes that will be suggested by the tool for the IVRS inventory management parameters to a quarterly frequency over the trial duration in order to allow for a practical implementation of the output.

Results and Analysis

From the treatment simulation, we obtain the demand envelope for each package type per label group. The cumulative demand is shown in Fig. 4 for the Placebo and Active 90 mg package types (US label group).

The black curve corresponds to the average demand, whereas the gray envelope shows the variability around this demand (minimum/maximum). At this stage, the variability shown is specific to the treatment (i.e., enrollment rate, dropout rate, visit schedule, randomization, dose titration). This means that the demand we observe here cannot be considered as the quantity to produce. Indeed, some overage will usually be needed to accommodate distribution and safety stocks at the different locations. The supply simulation will, therefore, allow for the determination of total overage requirements.

In this case, we observe that the variability in demand alone differs from one package type to another. The Placebo package cumulative demand presents a variability

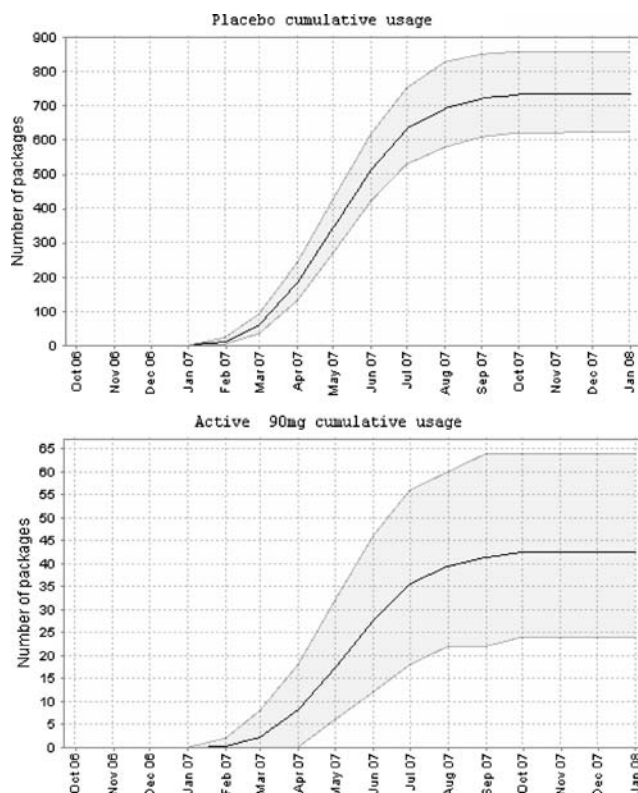


Fig. 4 Demand over time for package types Placebo and Active 90 mg (US label group)

of $\pm 16\%$, whereas the Active 90 mg package cumulative demand presents a variability of $\pm 47\%$.

This difference is due to the fact that the Placebo treatment group does not contain any uncertainty related to titration and weight ranges. Conversely, the Active 90 mg package is used only for patients with certain combinations of weight ranges and dose levels, which is highly variable.

For clarity purposes, we will focus on the results obtained for one package type (Active 90 mg) and one label group (US label group).

Preliminary observations show that for the Active 90 mg package type, the cumulative total average demand is 43 packages. Based on the study dispensing plan, one patient can receive up to six packages of Active 90 mg in total. Therefore, 43 packages correspond to potentially as few as seven patients.

When calculating the supply plan for this trial, if we were considering applying a global overage as a percentage added to the average demand, as it is often done without simulation, it could lead to insufficient quantities. For instance, if we decided to add 50% overage to the batch, we would produce 65 packages of Active 90 mg, which would potentially correspond in total to 10 patients. As there are eight sites in USA, it means that in practice, it would only be possible to cover about one patient per site on that dosage (assuming the worst case: that patients are on that dose level through all their visits). Given that the target of

enrollees in USA is 180 patients, this clearly appears to be an insufficient quantity as we do not know upfront at what sites specific patients will be. Therefore, in order to estimate the appropriate amount of overage for this package type as well as for the others—and in both label groups—the supply chain simulation will be crucial.

It is also important to note that the demand over time depends on the accuracy of the input provided for the simulation. Therefore, it is important to perform a sensitivity analysis. Although it is not presented here, it is interesting to investigate the impact of different enrollment and dropout rates, titration probabilities, etc. This is even more important when the trial is one of the first in a certain therapeutic area or indication and when historical data are not easily available.

The output of the overall simulation and optimization provides a set of supply scenarios (in this case 270), each containing different types of performance indicators:

- Number of stockout occurrences and corresponding percentage of missed visits (i.e., risk level)
- Shipment frequency and volume (i.e., distribution cost)
- Overage percentage and number of packages wasted (i.e., material cost).

In order to compare the different performance indicators, the relationship between two major costs (i.e., material, represented by the overage percentage, and distribution, represented by the shipment frequency) and risk is investigated. As shown in Fig. 5a, to maintain a constant service level, shipment frequency has to be increased when overage is decreased.

From these results, two major questions can be answered:

- What level of risk is acceptable for this trial?
- What combination of overage and shipment frequency should be selected?

In order to make an informed decision, the costs of material and shipments need to be considered.

Ideally, in order to perform a complete cost minimization, we should consider that risk also has a certain cost. However, in practice, that cost is very difficult to estimate, as it comprises not only tangible elements (e.g., cost of express shipment) but also intangible elements like the impact on patients and investigators' perception, as well as a potential study delay in worst cases.

Therefore, in practice, we will determine for each service level in the supply scenarios the optimal balance between the number of shipments and the overage level. This optimal balance is identified as the supply scenario generating the lowest cost. Figure 5b shows one example of the lowest (minimum) cost at constant service level. This graph shows that if the overage is not selected appropriately for a certain

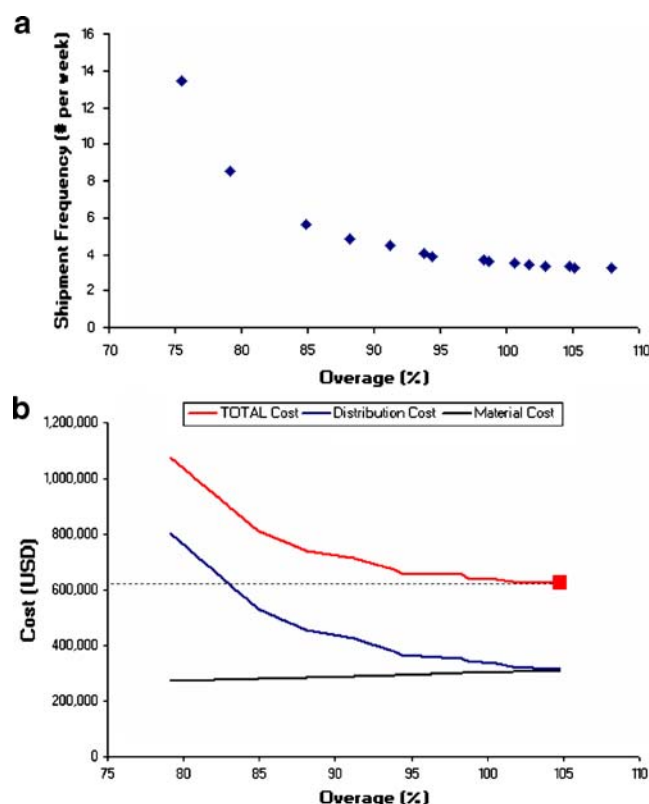


Fig. 5 **a** Shipment frequency as a function of overage for a constant risk level (0.034% of missed visits). **b** Material cost, distribution cost, and total cost as a function of overage for a constant risk level (0.034% of missed visits)

risk level, the cost could be significantly affected. In this example, when the overage is decreased below 105%, the cost of maintaining a constant service level through distribution increases. For instance, decreasing the overage level from 105% to 80% corresponds to a 72% increase in costs because of shipping costs. This type of finding demonstrates that waste minimization as an isolated target does not lead in all cases to cost minimization.

The different minimum costs as a function of risk are represented on Fig. 6.

The minimum cost increases when risk decreases, and two key observations can be made:

- After a certain point, if the risk level continues to be increased, the minimum cost does not decrease significantly.
- Conversely, to reach very low risk levels, the minimum cost is growing at an increasing rate.

This information is extremely valuable in defining the optimal supply strategy to pursue for a trial.

In this case study, the scenario corresponding to 0.038% of missed visits was selected and is highlighted on Fig. 6. The selection of a supply scenario can be based on predefined decision criteria, like the service level target

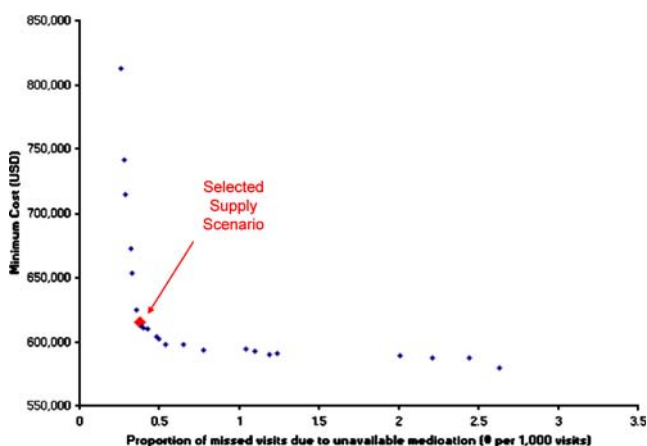


Fig. 6 Minimum cost as a function of risk level (in number of missed visits per 1,000 visits)

for the company. The choice of supply strategy will typically depend as well on the slope of the curve shown in Fig. 6, which provides an estimation of the incremental cost of decreasing risk.

Decreasing the proportion of missed visits from about three out of 1,000 to just less than one out of 1,000 does not correspond to a significant cost in this example (i.e., 0.31% increase). However, in order to decrease risk further, for example to 0.26 missed visits out of 1,000, the increase in cost is more substantial (i.e., 37% increase). In practice, the value associated with these percentages as well as the percentages themselves will depend on cost figures and trial design.

In addition, it is recommended to also consider the cost of resources that will increase when the risk is higher and typically when the overage is lower, since there will be more human intervention required.

It is also noted that the remaining risk can be further addressed by future supply strategy reevaluations.

A sample of the production plan and IVRS inventory management parameters corresponding to the selected supply scenario is shown in Fig. 7.

Figure 7a shows the quantities (Active 90 mg, US label group) and the time coverage of the two suggested productions, with variability. This information is particularly useful to identify any gap between two productions, especially if some material is expiring or when there are constraints relative to material availability. In Fig. 7b, the safety stock for one package type (Active 90 mg) and one site in the USA is represented over time. The selected scenario corresponds to a prediction window of 105 days.

The overall overage required for the study in this supply scenario is about 106%. This overage is spread very irregularly between the different package types, from 45% for Placebo to 310% for Active 90 mg. Typically, packages that are frequently dispensed like the Placebo one (50% of dispensing in this study design) require less overage than packages that are rarely assigned like the 90 mg dosage. Indeed, packages that are rarely assigned generate more uncertainty and correspond to smaller quantities which, therefore, require increased overage to stock all sites.

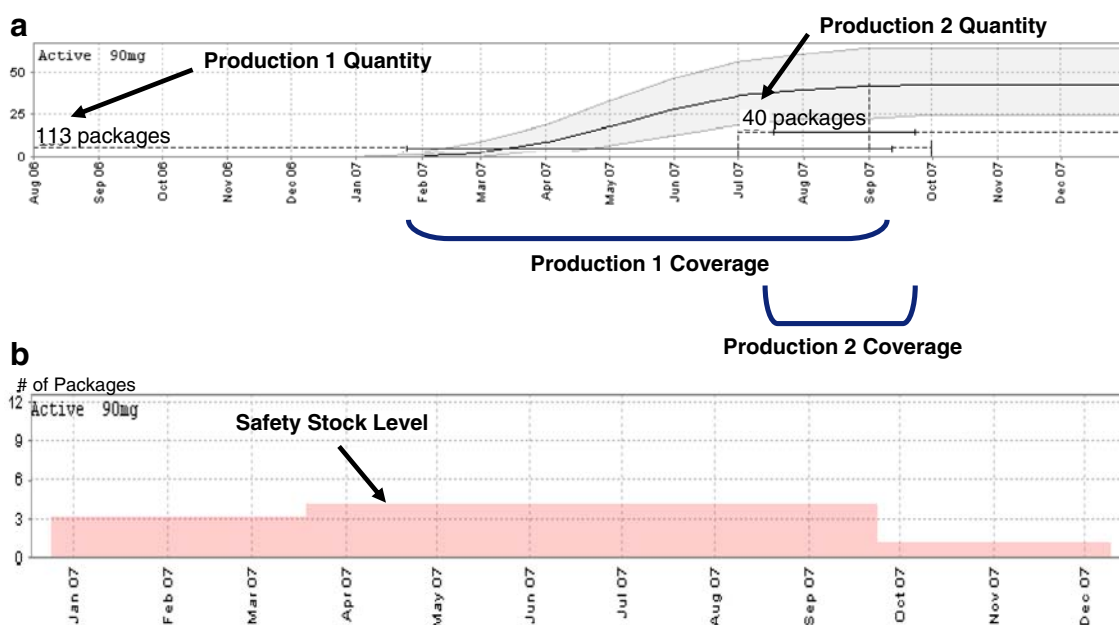


Fig. 7 **a** Production plan and production coverage (Active 90 mg, US material) for the selected supply scenario. **b** Safety stock over time (Active 90 mg, US site) for the selected supply scenario

In this trial, the level of variability is relatively high due to study-level competitive enrollment, study-level randomization [6], weight ranges, and dose titration, which leads to a relatively high percent overage for an acceptable level of risk.

Reevaluation with IVRS Actual Data

Simulation/Optimization Design

The trial supply chain is simulated again 7 months after the first patient visit to benefit from actual data. The simulation parameters are the same as for the pre-study forecast. The qualitative confidence level given to the assumptions made by the user in the balancing with actual data for the remaining of the trial is “medium.” This means that the reevaluated assumptions are the result of a balanced weighting between initial user assumptions and probabilities observed in the accumulated trial data.

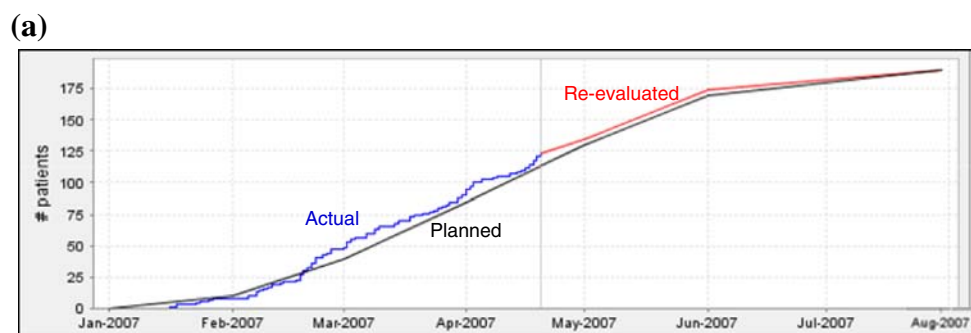
As a result of this reevaluation, the production plan and supply strategy are updated.

Results and Analysis

The initial input parameters are reevaluated using the Bayesian principle. An example (enrollment rate and titration probabilities) is shown in Fig. 8.

The reevaluated probabilities shown in Fig. 8b are based on the quantity of data available after 7 months and on the “medium” confidence level in the assumptions.

Fig. 8 **a** Enrollment rate reevaluation (USA). **b** Titration probabilities reevaluation (Male Stratum, Active treatment group, Visits 3 to 4)



(b)

From Visit 3 to Visit 4	Expected Probability	Number of Actual Observations	Re-evaluated Probability
Stay on Dose Level 3	32.5%	1	21.6%
Titrate Down to Dose Level 2	32.5%	3	22.9%
Titrate Down to Dose Level 1	32.5%	47	51.3%

The updated demand over time is shown in Fig. 9a. By utilizing the accumulated actual data, the variability for the remaining of the trial is reduced. In this example, the maximum cumulative demand increased from 64 to 84 for the Active 90 mg package type (US label group), mainly due to the reevaluated titration probabilities, which illustrates the importance of such reevaluation.

The same steps, as for the initial forecast, are followed for the selection of the optimal scenario. In this case, the quantities initially planned for the second production of Active 90 mg have to be increased. This is represented in Fig. 9b (US label group); at the time of reevaluation, 51 packages are left from the first production, and the suggested quantity for the second batch is now 62 instead of 40. In addition to changes to the production plan, updated safety stocks are provided for IVRS inventory management.

It is key to reevaluate the supply strategy on a regular basis and especially when certain milestones like enrollment closeout are reached. Indeed, as uncertainty is reduced over time, the forecasts will be more and more accurate, and production quantities as well as shipment frequency will typically be altered for an optimal result.

Operational and Implementation Aspects

Implementation considerations are key, as the approach described in this paper is based on decision modeling

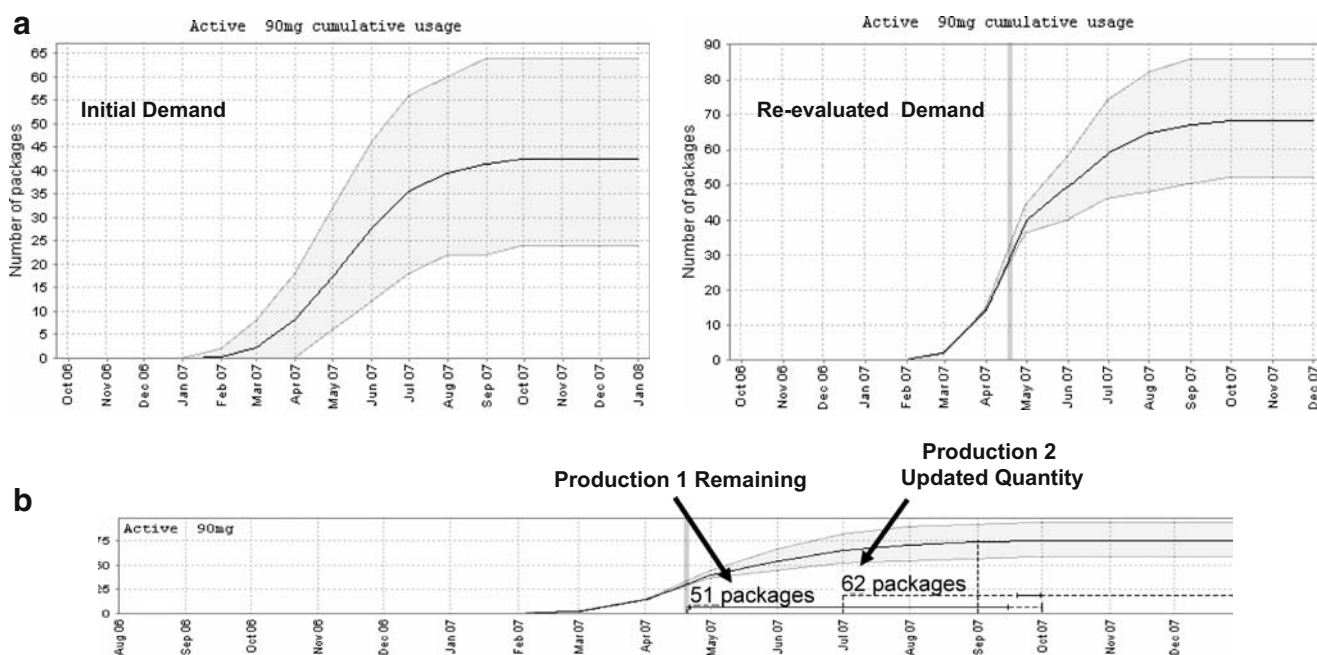


Fig. 9 **a** Initial and reevaluated demand over time for package type Active 90 mg (US label group). **b** Reevaluated production schedule for package type Active 90 mg (US label group)

techniques, which require expertise and training. This should be considered when going through the tasks of selecting a solution, deploying it, and then utilizing it as a core part of the clinical supplies management process.

Selection of a Solution

When evaluating the different solutions available on the market, it is critical to involve some people with a strong quantitative background or expertise, in conjunction with clinical supply experts, to assess the appropriateness and accuracy of modeling and algorithms. Indeed, the assumptions made to model trial conduct and supplies management have to be consistent with reality even when simplifications are needed to maintain an acceptable calculation time.

This constitutes a major difference compared to standard evaluation processes followed for nonmathematical IT applications like documentation and information management systems.

In addition, it is recommended to actually test the tool utilization in practice and even map the utilization process as some process steps (e.g., testing of various supply strategies) can have a huge impact on efficiency, user-friendliness, and ultimately effectiveness.

Implementation of a Solution

When this type of technology is directly used by clinical supply managers (as opposed to accessed through a consulting service), some implementation effort is required.

It is vital at that time and during at least the first months of utilization to benefit from comprehensive support provided by experts in both clinical supplies management and decision modeling.

In order to ensure a successful implementation, progressive training has to be provided to the users, starting with clinical supplies optimization principles and methodology.

It is also critical to determine how the tool will be integrated within the different existing processes and whether all clinical supply managers should use it or whether a specific role should be created, especially in the beginning.

In general, related appropriate goals and management metrics should be defined as well as supporting organization, processes, and functions [7] to ensure the success of this type of approach.

Utilization of a Solution

The use of a forecasting tool must be embedded in supply planning and inventory management processes. In particular, a standardized methodology as well as guidelines should be developed to (1) make robust assumptions and perform sensitivity analyses, (2) determine appropriate simulation parameters, and (3) analyze the output of the simulation in a consistent way. This includes clear decision criteria that support company-specific priorities. In addition, the output should be provided in a format that enables practical analysis and implementation. For instance, IVRS inventory management parameters, like minimum and maximum stock levels,

should be transferable in the IVR system in a convenient way. The optimal method is to feed them back automatically into the IVR system; however, if this is not feasible, it is important to consider some modeling adjustments that will enable a practical manual transfer throughout the trial (e.g., grouped changes at given times).

The tool interface must enable to minimize data entry time and match precisely the type of information that can be collected from clinical and other functions. When it is the case, entering a new trial in a forecasting tool is typically less time consuming than performing the same operation in Excel.

In order to achieve this, specific functionality can be created. For instance, titration probabilities need to be defined at each visit and transition between two dose levels; however, in practice, this information is almost impossible to obtain. If no facilitation is implemented in the tool, for a trial with 10 visits and five dose levels, clinical supply managers should determine and enter 50 probabilities: this is not only time consuming but also does not correspond to the type of information that can be obtained from clinical. A more realistic way of entering these data is to define an overall probability for patients to be on a certain dose level through a given study phase, combined with a qualitative estimate (low, medium, high) of transition probability within the same study phase. Other methods like rule-based titration can also be modeled to allow for a comprehensive definition of the trial design. In addition, in order to significantly accelerate data entry, it can be very useful to replace tables and multiple cells by graphical input, where for example clinical supply managers will draw on-screen the possible titration paths.

Conclusion

All the important steps of a clinical supply strategy definition were presented through a case study.

Clinical trial simulation alone is a powerful tool to assess demand and characterize cost and risk of a specific supply strategy. As described in this paper, the process of optimizing a site supply strategy (stocking levels, resupply quantities, and shipment frequency) with packaging overage and out of stock risk provides the clinical supply manager with the tools to balance cost and risk in a more effective and efficient way than traditional simulation-based approaches. In addition, the novel use of Bayesian principles applied to the reevaluation of study assumptions provides useful insight to assess the current supplies and more accurately target future production campaigns.

Use of such applications requires a solid foundation in the clinical supplies group of supply chain principles and planning, as well as support from decision sciences and statistics departments to ensure ongoing success and validation of the models.

The value created by this type of approach can be described through a balanced scorecard [8], with financial, internal process, customer satisfaction, and innovation and learning perspectives. Indeed, in addition to controlling risk and generating potentially substantial direct cost savings, it can enable accelerated trial execution, increase the control and agility of the clinical supplies organization, facilitate communication with clinical and other functions, and provide a robust foundation to develop clinical supplies management expertise.

Further technological developments could lead in the future to a fully automated cost optimization, with potentially sophisticated cost structures that would enable an even more comprehensive analysis in a reduced amount of time. The complete forecast could also include operations that occur before packaging, like drug product and active pharmaceutical ingredient manufacturing. Finally, continuous development based on current trends in clinical trial designs (e.g., adaptive designs, biologics specificities, etc.) should be undertaken in order to ensure that the forecasting tool can provide sustainable support for changing operations.

References

1. Slack N, Chambers S, Johnston R. Operations management. 5th ed. New York: Prentice Hall; 2007.
2. Byrom B. Managing the medication supply chain process using interactive response systems. *Life Sci Today*. 2002;3:16–8.
3. Dowlman N, et al. Optimizing the supply chain through trial simulation. *Appl Clin Trials*. 2004;13:40–6.
4. Banks J, et al. Discrete-event system simulation. 4th ed. New York: Prentice Hall; 2005.
5. Gelman A, et al. Bayesian data analysis. 2nd ed. New York: Chapman and Hall; 2004.
6. McEntegart D, O’Gorman B. The impact on supply logistics of different randomization and medication management strategies using IVR systems. *Pharm Eng*. 2005;25(5):36–47.
7. Rummler G, Brache A. Improving performance: how to manage the white space in the organization chart. San Francisco: Jossey-Bass; 1990.
8. Kaplan R, Norton D. The balanced scorecard: measures that drive performance. *Harvard Bus Rev*. 1992;70:79–80.